

DELBURNE AGCM, AB, CanadaWMO: **710140**Lat: **52.1828N**Lon: **113.1757W**Elev: **2917**StdP: **13.21**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-19.7	-14.6	-26.2	1.4	-19.5	-20.6	2.0	-14.1	23.1	24.0	20.7	22.7	5.3	150	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	82.0	62.1	78.9	60.7	75.9	59.6	65.5	76.2	63.4	74.4	61.7	72.4	6.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.6	91.4	70.5	59.4	84.2	68.4	57.3	78.0	66.4	32.0	76.2	30.3	74.7	29.0	72.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.0	17.5	15.1	DB	-25.9	88.3	4.6	4.4	-29.2	91.5	-31.9	94.0	-34.5	96.5	-37.8	99.7	(4)
(5)			WB	-26.4	69.2	4.4	2.6	-29.5	71.1	-32.1	72.6	-34.5	74.1	-37.7	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.7	17.1	16.1	26.3	38.3	50.9	57.6	62.3	60.9	52.5	40.0	26.1	14.5	(6)	
	DBStd	20.43	15.21	14.65	13.36	9.45	8.06	5.52	4.98	5.72	8.52	9.15	12.80	14.30	(7)	
	HDD50	5380	1019	950	736	361	87	8	0	2	69	328	719	1101	(8)	
	HDD65	9672	1484	1370	1201	800	439	228	113	149	381	774	1167	1566	(9)	
	CDD50	1247	0	0	0	11	115	236	381	341	143	18	2	0	(10)	
	CDD65	70	0	0	0	0	3	7	30	24	6	0	0	0	(11)	
	CDH74	1059	0	0	0	5	74	104	381	348	143	4	0	0	(12)	
	CDH80	193	0	0	0	0	4	11	75	74	29	0	0	0	(13)	
	WSAvg	6.6	6.4	6.7	6.8	7.4	7.2	6.5	5.7	5.3	5.9	7.1	6.8	6.8	(14)	
	PrecAvg	18.5	0.9	0.6	0.8	1.1	2.1	3.5	3.5	2.3	1.4	0.9	0.8	0.7	(15)	
(16) Precipitation	PrecMax	22.6	2.9	1.5	1.8	2.6	4.1	5.7	7.0	4.4	3.0	2.2	2.4	1.3	(16)	
	PrecMin	13.4	0.1	0.1	0.2	0.2	0.7	1.0	1.0	0.7	0.1	0.2	0.2	0.2	(17)	
	PrecStd	2.5	0.6	0.3	0.4	0.6	0.9	1.1	1.4	0.8	0.8	0.5	0.5	0.4	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	47.0	57.5	71.2	80.2	80.8	86.0	86.2	84.2	72.0	60.6	46.7	(19)	
		MCWB	38.4	38.5	43.5	49.1	54.9	59.6	65.2	61.2	57.4	52.7	46.3	37.5	(20)	
	2%	DB	44.1	42.3	51.3	65.3	75.8	77.0	82.1	81.3	78.3	63.9	52.4	41.3	(21)	
		MCWB	36.6	35.1	40.0	47.3	53.5	59.4	64.1	62.0	57.4	49.3	42.0	34.2	(22)	
	5%	DB	40.9	38.4	46.8	59.1	71.2	73.6	78.2	77.9	73.2	58.8	45.9	38.0	(23)	
		MCWB	34.4	32.6	37.7	43.5	52.1	57.5	62.9	61.7	55.9	46.2	37.8	31.9	(24)	
	10%	DB	37.3	35.1	42.4	53.8	66.6	70.1	74.8	74.1	68.0	54.0	41.6	33.6	(25)	
		MCWB	31.9	30.6	35.4	41.0	50.3	55.9	61.4	60.8	53.7	43.8	35.2	28.9	(26)	
	0.4%	WB	39.2	38.9	44.6	50.6	57.0	64.5	69.3	67.3	61.0	54.0	47.2	38.4	(27)	
		MCDB	46.3	46.3	55.9	69.7	74.8	75.4	79.4	77.5	76.5	68.6	58.8	45.1	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.0	35.5	40.9	47.3	54.8	61.6	66.2	65.1	58.8	50.5	42.0	34.9	(29)	
		MCDB	43.6	41.7	49.9	64.2	71.1	73.3	76.7	76.0	73.8	61.5	50.8	41.3	(30)	
	5%	WB	34.7	33.0	38.0	44.5	53.2	59.4	64.1	63.2	56.7	47.4	38.7	32.1	(31)	
		MCDB	40.5	37.6	46.2	56.7	68.4	69.9	75.0	74.3	70.9	57.7	45.0	37.0	(32)	
	10%	WB	32.0	30.8	35.5	41.7	51.4	57.4	62.4	61.2	54.7	44.5	35.7	29.1	(33)	
		MCDB	37.1	35.0	41.7	52.2	64.6	67.4	72.7	72.1	67.1	52.6	40.4	33.4	(34)	
	5% DB	MDBR	15.9	17.1	17.6	19.7	21.4	19.8	21.0	21.7	21.7	18.8	15.7	15.1	(35)	
		MCDBR	16.5	19.9	22.3	29.5	28.5	25.4	26.6	28.2	31.0	26.0	22.0	18.1	(36)	
		MCWBR	11.4	14.5	13.3	14.7	11.9	11.0	12.2	12.8	13.9	14.0	13.3	13.2	(37)	
		MCWBR	16.1	19.0	21.5	27.6	24.8	22.1	24.0	24.6	28.2	25.5	21.0	17.6	(38)	
	5% WB	MCWBR	11.6	14.2	13.4	14.1	10.9	10.8	12.2	12.5	13.3	13.9	13.2	13.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.262	0.269	0.286	0.322	0.339	0.346	0.346	0.359	0.309	0.281	0.252	0.243	(40)	
	taud		2.311	2.390	2.436	2.377	2.385	2.395	2.419	2.362	2.476	2.535	2.473	2.332	(41)	
	Ebn at Noon		238	275	292	291	290	287	285	275	279	269	247	227	(42)	
	Edh at Noon		19	24	28	34	36	36	34	34	27	21	16	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		344	675	1118	1499	1824	1935	2038	1660	1157	693	358	258	(44)	
	RadStd		30	56	86	134	116	121	105	115	124	81	35	25	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page